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Detection device

Abstract

A detection device includes a substrate, a plurality of photodiodes that are arranged in a detection region of the substrate, a plurality of first terminals that are arranged in a first direction in a peripheral region outside the detection region of the substrate, an insulating film that covers the first terminals, and an anisotropic conductive film that is located above the insulating film, and covers the first terminals.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation of International Patent Application No. PCT/JP2021/021372 filed on Jun. 4, 2021 which designates the United States, incorporated herein by reference, and which claims the benefit of priority from Japanese Patent Application No. 2020-107100 filed on Jun. 22, 2020, incorporated herein by reference.

BACKGROUND

1. Technical Field

(1) The present invention relates to an electronic apparatus, and in particular, to a detection device.

2. Description of the Related Art

(2) A liquid crystal display device of Japanese Patent Application Laid-open Publication No. 2010-277378 A includes a plurality of optical sensors. The optical sensors each include a photodiode. The photodiode converts light emitted thereto into a signal (electric charge). The optical sensors are generally arranged in a matrix having a row-column configuration. The optical sensors arranged in a matrix having a row-column configuration are used in detection devices, for example, as biometric sensors, such as fingerprint sensors and vein sensors, that detect biometric information. A substrate of the optical sensors is provided there on with a plurality of terminals for electrically coupling to external circuitry. The terminals are coupled to a wiring substrate, such as a flexible printed circuit board, and integrated circuits (ICs).

(3) The optical sensors including a plurality of photodiodes are required to be improved in coupling reliability of the terminals.

SUMMARY

(4) A detection device according to an embodiment of the present disclosure includes a substrate, a plurality of photodiodes that are arranged in a detection region of the substrate, a plurality of first terminals that are arranged in a first direction in a peripheral region outside the detection region of the substrate, an insulating film that covers the first terminals, and an anisotropic conductive film that is located above the insulating film, and covers the first terminals. Each of the first terminals comprises, between the substrate and the insulating film, a first metal layer, a second metal layer that is stacked above the first metal layer with a first interlayer insulating film interposed between the first metal layer and the second metal layer, a third metal layer that is stacked above the second metal layer so as to be in contact with the second metal layer, and a first light-transmitting conductive layer that is stacked above the third metal layer so as to be in contact with the third metal layer, the insulating film has an opening that exposes the first light-transmitting conductive layer in a region overlapping each of the first terminals, and the anisotropic conductive film is in direct contact with a side surface of the insulating film forming the opening and with the first light-transmitting conductive layer overlapping the opening.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1**A is a sectional view illustrating a schematic sectional configuration of a detection apparatus having an illumination device, the detection apparatus including a detection device according to a first embodiment;

(2) FIG. **1**B is a sectional view illustrating a schematic sectional configuration of the detection apparatus having an illumination device, the detection apparatus including the detection device according to a first modification;

(3) FIG. **1**C is a sectional view illustrating a schematic sectional configuration of the detection apparatus having an illumination device, the detection apparatus including the detection device according to a second modification;

(4) FIG. **1**D is a sectional view illustrating a schematic sectional configuration of the detection apparatus having an illumination device, the detection apparatus including the detection device according to a third modification;

(5) FIG. **2** is a plan view illustrating the detection device according to the first embodiment;

(6) FIG. **3** is a block diagram illustrating a configuration example of the detection device according to the first embodiment;

(7) FIG. **4** is a circuit diagram illustrating a detection element;
(8) FIG. **5** is a plan view illustrating an array substrate on which the detection element is formed;
(9) FIG. **6** is a plan view illustrating the detection element on the array substrate;
(10) FIG. **7** is a VII-VII′ sectional view of FIG. **5**;
(11) FIG. **8** is a plan view schematically illustrating a configuration of a terminal portion of the detection device according to the first embodiment;
(12) FIG. **9** is a plan view illustrating a magnified view of a first terminal;
(13) FIG. **10** is a X-X′ sectional view of FIG. **9**;
(14) FIG. **11** is an XI-XI′ sectional view of FIG. **9**;
(15) FIG. **12** is a sectional view illustrating a structure example of bonding between the array substrate and a wiring substrate according to the first embodiment;
(16) FIG. **13** is a XIII-XIII′ sectional view of FIG. **8**;
(17) FIG. **14** is a sectional view schematically illustrating a first terminal according to a second embodiment;
(18) FIG. **15** is a sectional view schematically illustrating a first terminal according to a third embodiment;
(19) FIG. **16** is a plan view schematically illustrating a configuration of a terminal portion of a detection device according to a fourth embodiment;
(20) FIG. **17** is a plan view illustrating a magnified view of the first terminal according to the fourth embodiment;
(21) FIG. **18** is an XVIII-XVIII′ sectional view of FIG. **17**;
(22) FIG. **19** is an explanatory view for explaining an arrangement relation among the first terminals, a plurality of second terminals, outer edge wiring, and insulating films;
(23) FIG. **20** is a XX-XX′ sectional view of FIG. **19**;
(24) FIG. **21** is an explanatory view for explaining an arrangement relation among the first terminals, the second terminals, the outer edge wiring, and the insulating films according to a fourth modification of the fourth embodiment;
(25) FIG. **22** is a XXII-XXII′ sectional view of FIG. **21**;
(26) FIG. **23** is an explanatory view for explaining an arrangement relation among the first terminals, the second terminals, the outer edge wiring, and the insulating films according to a fifth modification of the fourth embodiment; and
(27) FIG. **24** is a XXIV-XXIV′ sectional view of FIG. **23**.

DETAILED DESCRIPTION

(28) The following describes modes (embodiments) for carrying out the present invention in detail with reference to the drawings. The present invention is not limited to the description of the embodiments to be given below. Components to be described below include those easily conceivable by those skilled in the art or those substantially identical thereto. In addition, the components to be described below can be combined as appropriate. What is disclosed herein is merely an example, and the present invention naturally encompasses appropriate modifications easily conceivable by those skilled in the art while maintaining the gist of the invention. To further clarify the description, the drawings may schematically illustrate, for example, widths, thicknesses, and shapes of various parts as compared with actual aspects thereof. However, they are merely examples, and interpretation of the present invention is not limited thereto. The same component as that described with reference to an already mentioned drawing is denoted by the same reference numeral through the description and the drawings, and detailed description thereof may not be repeated where appropriate.
(29) In the present specification and claims, in expressing an aspect of disposing a second structure above a first structure, a case of simply expressing “above” includes both a case of disposing the second structure immediately above the first structure so as to contact the first structure and a case of disposing the second structure above the first structure with a a third structure interposed

therebetween, unless otherwise specified.

First Embodiment

(30) FIG. **1**A is a sectional view illustrating a schematic sectional configuration of a detection apparatus having an illumination device, the detection apparatus including a detection device according to a first embodiment. FIG. **1**B is a sectional view illustrating a schematic sectional configuration of the detection apparatus having an illumination device, the detection apparatus including the detection device according to a first modification. FIG. **1**C is a sectional view illustrating a schematic sectional configuration of the detection apparatus having an illumination device, the detection apparatus including the detection device according to a second modification. FIG. **1**D is a sectional view illustrating a schematic sectional configuration of the detection apparatus having an illumination device, the detection apparatus including the detection device according to a third modification.

(31) As illustrated in FIG. **1**A, a detection apparatus **120** having an illumination device includes a detection device **1** and an illumination device **121**. The detection device **1** includes an array substrate **2**, an adhesive layer **125**, and a cover member **122**. That is, the array substrate **2**, the adhesive layer **125**, and the cover member **122** are stacked in this order in a direction orthogonal to a surface of the array substrate **2**. As will be describe later, the cover member **122** of the detection device **1** may be replaced with the illumination device **121**.

(32) As illustrated in FIG. **1**A, the illumination device **121** may be, for example, what is called a side light-type front light that uses the cover member **122** as a light guide plate provided in a position corresponding a detection region AA of the detection device **1** and includes a plurality of light sources **123** arranged at one end or both ends of the cover member **122**. That is, the cover member **122** has a light-emitting surface **121***a* for emitting light, and serves as one component of the illumination device **121**. The illumination device **121** emits light L**1** from the light-emitting surface **121***a* of the cover member **122** toward a finger Fg that serves as a detection target. For example, light-emitting diodes (LEDs) for emitting light in a predetermined color are used as the light sources.

(33) As illustrated in FIG. **1**B, the illumination device **121** may include the light sources (for example, LEDs) provided directly below the detection region AA of the detection device **1**. The illumination device **121** including the light sources also serves as the cover member **122**.

(34) The illumination device **121** is not limited to the example of FIG. **1**B. As illustrated in FIG. **1**C, the illumination device **121** may be provided on a lateral side or an upper side of the cover member **122**, and may emit the light L**1** to the finger Fg from the lateral side or the upper side of the finger Fg.

(35) Furthermore, as illustrated in FIG. **1**D, the illumination device **121** may be what is called a direct-type backlight that includes the light sources (for example, LEDs) provided in the detection region of the detection device **1**.

(36) The light L**1** emitted from the illumination device **121** is reflected as light L**2** by the finger Fg serving as the detection target. The detection device **1** detects the light L**2** reflected by the finger Fg to detect asperities (such as a fingerprint) on a surface of the finger Fg. The detection device **1** may further detect information on a living body by detecting the light L**2** reflected in the finger Fg, in addition to detecting the fingerprint. Examples of the information on the living body include a blood vessel image, pulsation, and a pulse wave of, for example, a vein. The color of the light L**1** from the illumination device **121** may be varied according to the detection target.

(37) The cover member **122** is a member for protecting the array substrate **2**, and covers the array substrate **2**. The illumination device **121** may have a structure to serve also as the cover member **122**, as described above. In the structures illustrated in FIGS. **1**C and **1**D in which the cover member **122** is separate from the illumination device **121**, the cover member **122** is, for example, a glass substrate. The cover member **122** is not limited to the glass substrate, and may be a resin substrate, for example. The cover member **122** may be omitted. In that case, the surface of the array

substrate **2** is provided with a protective layer of, for example, an insulating film, and the finger Fg contacts the protective layer of the detection device **1**.
(38) The detection apparatus **120** having an illumination device may be provided with a display panel instead of the illumination device **121**, as illustrated in FIG. **1**B. The display panel may be, for example, an organic electroluminescent (EL) (organic light-emitting diode (OLED)) display panel or an inorganic EL (micro-LED or mini-LED) display panel. Alternatively, the display panel may be a liquid crystal display (LCD) panel using liquid crystal elements as display elements or an electrophoretic display (EPD) panel using electrophoretic elements as the display elements. Even in this case, the fingerprint of the finger Fg and the information on the living body can be detected based on the light L**2** obtained by reflecting display light (light L**1**) emitted from the display panel against the finger Fg.
(39) FIG. **2** is a plan view illustrating the detection device according to the first embodiment. A first direction Dx illustrated in FIG. **2** and the subsequent drawings is one direction in a plane parallel to a substrate **21**. A second direction Dy is one direction in the plane parallel to the substrate **21**, and is a direction orthogonal to the first direction Dx. The second direction Dy may non-orthogonally intersect the first direction Dx. A third direction Dz is a direction orthogonal to the first direction Dx and the second direction Dy, and is a direction normal to the substrate **21**.
(40) As illustrated in FIG. **2**, the detection device **1** includes the array substrate **2** (substrate **21**), a sensor unit **10**, a scan line drive circuit **15** (first scan line drive circuit **15**A and second scan line drive circuit **15**B), a signal line selection circuit **16**, a detection circuit **48**, a control circuit **102**, and a power supply circuit **103**.
(41) The substrate **21** is electrically coupled to a control substrate **101** through a wiring substrate **110**. The wiring substrate **110** is, for example, a flexible printed circuit board or a rigid circuit board. The wiring substrate **110** is provided with the detection circuit **48**. The control substrate **101** is provided with the control circuit **102** and the power supply circuit **103**. The control circuit **102** is, for example, a field-programmable gate array (FPGA). The control circuit **102** supplies control signals to the sensor unit **10**, the scan line drive circuit **15**, and the signal line selection circuit **16** to control an operation of the sensor unit **10**. The power supply circuit **103** supplies voltage signals including, for example, a power supply potential VDD and a reference potential VCOM (refer to FIG. **4**) to the sensor unit **10**, the scan line drive circuit **15**, and the signal line selection circuit **16**. In the present embodiment, the case is exemplified where the detection circuit **48** is disposed on the wiring substrate **110**, but the present invention is not limited to this case. The detection circuit **48** may be disposed on the substrate **21**.
(42) The substrate **21** has the detection region AA and a peripheral region GA. The detection region AA and the peripheral region GA extend in planar directions parallel to the substrate **21**. Elements (detection elements **3**) of the sensor unit **10** are provided in the detection region AA. The peripheral region GA is a region outside the detection region AA, and is a region not provided with the elements (detection elements **3**). That is, the peripheral region GA is a region between the outer periphery of the detection region AA and the outer edges of the substrate **21**.
(43) The scan line drive circuit **15** and the signal line selection circuit **16** are provided in the peripheral region GA. The scan line drive circuit **15** is provided in a region extending along the second direction Dy in the peripheral region GA. The scan line drive circuit **15** includes the first scan line drive circuit **15**A and the second scan line drive circuit **15**B. The first scan line drive circuit **15**A and the second scan line drive circuit **15**B are arranged adjacent to each other in the first direction Dx with the detection region AA interposed therebetween. In the following description, the first scan line drive circuit **15**A and the second scan line drive circuit **15**B are simply called the scan line drive circuit **15** when they need not be distinguished from each other. The signal line selection circuit **16** is provided in a region extending along the first direction Dx in the peripheral region GA, and is provided between the sensor unit **10** and the detection circuit **48**.
(44) Each of the detection elements **3** of the sensor unit **10** is an optical sensor including a

photodiode **30** as a sensor element. The photodiode **30** is a photoelectric conversion element, and outputs an electrical signal corresponding to light irradiating each of the photodiodes **30**. More specifically, the photodiode **30** is a positive-intrinsic-negative (PIN) photodiode. The photodiode **30** may be paraphrased as an organic photodiode (OPD). The detection elements **3** are arranged in a matrix having a row-column configuration in the detection region AA. The photodiode **30** included in each of the detection elements **3** performs the detection according to gate drive signals (for example, a reset control signal RST and a read control signal RD) supplied from the scan line drive circuit **15**. Each of the photodiodes **30** outputs the electrical signal corresponding to the light irradiating the photodiode **30** as a detection signal Vdet to the signal line selection circuit **16**. The detection device **1** detects the information on the living body based on the detection signals Vdet received from the photodiodes **30**.
(45) FIG. **3** is a block diagram illustrating a configuration example of the detection device according to the first embodiment. As illustrated in FIG. **3**, the detection device **1** further includes a detection control circuit **11** and a detector **40**. The control circuit **102** includes one, some, or all functions of the detection control circuit **11**. The control circuit **102** also includes one, some, or all functions of the detector **40** other than those of the detection circuit **48**.
(46) The detection control circuit **11** is a circuit that supplies respective control signals to the scan line drive circuit **15**, the signal line selection circuit **16**, and the detector **40** to control operations of these components. The detection control circuit **11** supplies various control signals including, for example, a start signal STV and a clock signal CK to the scan line drive circuit **15**. The detection control circuit **11** also supplies various control signals including, for example, a selection signal ASW to the signal line selection circuit **16**.
(47) The scan line drive circuit **15** is a circuit that drives a plurality of scan lines (read control scan line GLrd and reset control scan line GLrst (refer to FIG. **4**)) based on the various control signals. For example, the first scan line drive circuit **15**A (refer to FIG. **2**) scans one of the read control scan line GLrd and the reset control scan line GLrst. The second scan line drive circuit **15**B (refer to FIG. **2**) scans the other of the read control scan line GLrd and the reset control scan line GLrst. The scan line drive circuit **15** sequentially or simultaneously selects the scan lines, and supplies the gate drive signals (for example, the reset control signals RST and the read control signals RD) to the selected scan lines. Through this operation, the scan line drive circuit **15** selects the photodiodes **30** coupled to the scan lines.
(48) The signal line selection circuit **16** is a switch circuit that sequentially or simultaneously selects a plurality of output signal lines SL (refer to FIG. **4**). The signal line selection circuit **16** is, for example, a multiplexer. The signal line selection circuit **16** couples the selected output signal lines SL to the detection circuit **48** based on the selection signal ASW supplied from the detection control circuit **11**. Through this operation, the signal line selection circuit **16** outputs the detection signals Vdet of the photodiodes **30** to the detector **40**.
(49) The detector **40** includes the detection circuit **48**, a signal processing circuit **44**, a coordinate extraction circuit **45**, a storage circuit **46**, and a detection timing control circuit **47**. The detection timing control circuit **47** performs control to cause the detection circuit **48**, the signal processing circuit **44**, and the coordinate extraction circuit **45** to operate in synchronization with one another based on a control signal supplied from the detection control circuit **11**.
(50) The detection circuit **48** is, for example, an analog front-end (AFE) circuit. The detection circuit **48** is a signal processing circuit having functions of at least a detection signal amplifying circuit **42** and an analog-to-digital (A/D) conversion circuit **43**. The detection signal amplifying circuit **42** amplifies the detection signal Vdet, and is an integration circuit, for example. The A/D conversion circuit **43** converts an analog signal output from the detection signal amplifying circuit **42** into a digital signal.
(51) The signal processing circuit **44** is a logic circuit that detects a predetermined physical quantity received by the sensor unit **10** based on output signals of the detection circuit **48**. The

signal processing circuit **44** can detect asperities on the surface of the finger Fg or a palm based on the signals from the detection circuit **48** when the finger Fg is in contact with or in proximity to a detection surface. The signal processing circuit **44** may detect the information on the living body based on the signals from the detection circuit **48**. Examples of the information on the living body include a blood vessel image, a pulse wave, pulsation, and a blood oxygen saturation level of the finger Fg or the palm. That is, the detection device **1** may be configured as a fingerprint sensor for detecting the fingerprint or a vein sensor for detecting a vascular pattern of, for example, veins.

(52) The storage circuit **46** temporarily stores therein signals calculated by the signal processing circuit **44**. The storage circuit **46** may be, for example, a random-access memory (RAM) or a register circuit.

(53) The coordinate extraction circuit **45** is a logic circuit that obtains detected coordinates of the asperities on the surface of the finger Fg or the like when the contact or proximity of the finger Fg is detected by the signal processing circuit **44**. The coordinate extraction circuit **45** is the logic circuit that also obtains detected coordinates of blood vessels of the finger Fg or the palm. The coordinate extraction circuit **45** combines the detection signals Vdet output from the respective detection elements **3** of the sensor unit **10** to generate two-dimensional information representing a shape of the asperities on the surface of the finger Fg or the like. The coordinate extraction circuit **45** may output the detection signals Vdet as sensor outputs Vo instead of calculating the detected coordinates.

(54) The following describes a circuit configuration example of the detection device **1**. FIG. **4** is a circuit diagram illustrating the detection element. As illustrated in FIG. **4**, the detection element **3** includes the photodiode **30**, a reset transistor Mrst, a read transistor Mrd, and a source follower transistor Msf. The reset transistor Mrst, the read transistor Mrd, and the source follower transistor Msf are provided correspondingly to each of the photodiodes **30**. The reset transistor Mrst, the read transistor Mrd, and the source follower transistor Msf are each constituted by an n-type thin-film transistor (TFT). However, each of the transistors is not limited thereto, and may be constituted by a p-type TFT.

(55) The reference potential VCOM is applied to an anode of the photodiode **30**. A cathode of the photodiode **30** is coupled to a node N**1**. The node N**1** is coupled to a capacitive element Cs, one of the source and the drain of the reset transistor Mrst, and the gate of the source follower transistor Msf. The node N**1** further has parasitic capacitance Cp. When light is incident on the photodiode **30**, a signal (electric charge) output from the photodiode **30** is stored in the capacitive element Cs. The capacitive element Cs is a capacitor formed between an upper electrode **34** and a lower electrode **35** (refer to FIG. **7**) that are coupled to the photodiode **30**. The parasitic capacitance Cp is capacitance added to the capacitive element Cs, and is capacitance generated among various types of wiring and electrodes provided on the array substrate **2**.

(56) The gate of the reset transistor Mrst is coupled to the reset control scan line GLrst. The other of the source and the drain of the reset transistor Mrst is coupled to a reset signal line SLrst, and is supplied with a reset potential Vrst. When the reset transistor Mrst is turned on (into a conduction state) in response to the reset control signal RST, the potential of the node N**1** is reset to the reset potential Vrst. The reference potential VCOM is lower than the reset potential Vrst, and the photodiode **30** is driven in a reverse bias state.

(57) The source follower transistor Msf is coupled between a terminal supplied with the power supply potential VDD and the read transistor Mrd (node N**2**). The gate of the source follower transistor Msf is coupled to the node N**1**. The gate of the source follower transistor Msf is supplied with a signal (electric charge) generated by the photodiode **30**. This operation causes the source follower transistor Msf to output a voltage signal corresponding to the signal (electric charge) generated by the photodiode **30** to the read transistor Mrd.

(58) The read transistor Mrd is coupled between the source of the source follower transistor Msf (node N**2**) and a corresponding one of the output signal lines SL (node N**3**). The gate of the read

transistor Mrd is coupled to the read control scan line GLrd. When the read transistor Mrd is turned on in response to the read control signal RD, the signal output from the source follower transistor Msf, that is, the voltage signal corresponding to the signal (electric charge) generated by the photodiode **30** is output as the detection signal Vdet to the output signal line SL.

(59) In the example illustrated in FIG. **4**, the reset transistor Mrst and the read transistor Mrd each have what is called a double-gate structure configured by coupling two transistors in series. However, the reset transistor Mrst and the read transistor Mrd are not limited to this structure, and may have a single-gate structure, or a multi-gate structure including three or more transistors coupled in series. The circuit of each of the detection elements **3** is not limited to the configuration including the three transistors of the reset transistor Mrst, the source follower transistor Msf, and the read transistor Mrd. The detection element **3** may include two transistors, or four or more transistors.

(60) The following describes a planar configuration of the detection element **3**. FIG. **5** is a plan view illustrating the array substrate on which the detection element is formed. FIG. **6** is a plan view illustrating the detection element on the array substrate. FIG. **5** is a plan view schematically illustrating a portion of the detection element **3**, that is, a portion thereof except members above the photodiode **30**. FIG. **5** illustrates the lower electrode **35** and the photodiode **30** with long dashed double-short dashed lines.

(61) As illustrated in FIG. **5**, the reset control scan lines GLrst each extend in the first direction Dx, and are separately arranged in the second direction Dy. The output signal lines SL each extend in the second direction Dy, and are separately arranged in the first direction Dx. The photodiode **30** of the detection element **3** is provided in a region surrounded by two of the reset control scan lines GLrst adjacent in the second direction Dy and two of the output signal lines SL adjacent in the first direction Dx.

(62) The detection element **3** further includes the read control scan line GLrd and two signal lines (power supply signal line SLsf and reset signal line SLrst). The read control scan line GLrd extends in the first direction Dx, and is arranged side by side with the reset control scan line GLrst in the second direction Dy. Each of the power supply signal line SLsf and the reset signal line SLrst extends in the second direction Dy, and is arranged side by side with the output signal line SL in the first direction Dx.

(63) As illustrated in FIG. **5**, the reset transistor Mrst of the detection element **3** includes a first semiconductor layer **61**, a source electrode **62**, a drain electrode **63**, and gate electrodes **64**. One end of the first semiconductor layer **61** is coupled to the reset signal line SLrst. The other end of the first semiconductor layer **61** is coupled to coupling wiring SLcn. A portion of the reset signal line SLrst coupled to the first semiconductor layer **61** serves as the source electrode **62**, and a portion of the coupling wiring SLcn coupled to the first semiconductor layer **61** serves as the drain electrode **63**.

(64) The gate electrodes **64** face the first semiconductor layer **61**. More specifically, the reset control scan line GLrst is provided with two branches branching in the second direction Dy, and the first semiconductor layer **61** extends in the first direction Dx and intersects the two branches of the reset control scan line GLrst. Channel regions are formed at portions of the first semiconductor layer **61** overlapping the two branches of the reset control scan line GLrst, and portions of the two branches of the reset control scan line GLrst that overlap the first semiconductor layer **61** serve as the gate electrodes **64**. Thus, the reset transistor Mrst is configured as a double-gate structure in which the two gate electrodes **64** are provided so as to overlap the first semiconductor layer **61**.

(65) The source follower transistor Msf of the detection element **3** includes a second semiconductor layer **65**, a source electrode **67**, and a gate electrode **68**. One end of the second semiconductor layer **65** is coupled to the power supply signal line SLsf through a coupling portion SLsfa. The other end of the second semiconductor layer **65** is coupled to the read transistor Mrd. A portion of the coupling portion SLsfa coupled to the second semiconductor layer **65** serves as the source electrode

67.
(66) One end of the gate electrode **68** is coupled to the coupling wiring SLcn through a contact hole. The second semiconductor layer **65** intersects the gate electrode **68**. A channel region is formed at a portion of the second semiconductor layer **65** intersecting the gate electrode **68**. The source follower transistor Msf is configured as a single-gate structure in which the one gate electrode **68** is provided so as to overlap the second semiconductor layer **65**. The reset transistor Mrst is electrically coupled to the gate of the source follower transistor Msf through the coupling wiring SLcn.
(67) The coupling wiring SLcn is disposed between the power supply signal line SLsf and the output signal line SL adjacent to each other in the first direction Dx. The coupling wiring SLcn includes a portion that is coupled to the reset transistor Mrst and extends in the first direction Dx, and a portion that is coupled to the source follower transistor Msf and extends in the second direction Dy. The cathode (n-type semiconductor layer **33**) of the photodiode **30** of the detection element **3** is coupled to the coupling wiring SLcn through a contact hole H**2**. This configuration electrically couples the cathode (n-type semiconductor layer **33**) of the photodiode **30** to the reset transistor Mrst and the source follower transistor Msf through the coupling wiring SLcn.
(68) The read transistor Mrd includes the second semiconductor layer **65**, a drain electrode **72**, and gate electrodes **74**. The second semiconductor layer **65** of the read transistor Mrd is formed of a semiconductor layer integrated with the second semiconductor layer **65** of the source follower transistor Msf. In other words, the read transistor Mrd and the source follower transistor Msf include the common second semiconductor layer **65**. The other end of the second semiconductor layer **65** of the read transistor Mrd is coupled to the output signal line SL through a coupling portion SLa. In other words, a portion of the coupling portion SLa coupled to the second semiconductor layer **65** serves as the drain electrode **72**.
(69) The read control scan line GLrd is coupled to a branch that is adjacent thereto in the second direction Dy and extends in the first direction Dx. The second semiconductor layer **65** intersects the read control scan line GLrd and the branch. Portions of the read control scan line GLrd and the branch thereof that overlap the second semiconductor layer **65** serve as the gate electrodes **74**. Thus, the read transistor Mrd is configured as a double-gate structure in which the two gate electrodes **74** are provided so as to overlap the second semiconductor layer **65**.
(70) In the present embodiment, the second semiconductor layer **65** is arranged adjacent in the first direction Dx to the output signal line SL, and the second semiconductor layer **65** and the output signal line SL extend in the second direction Dy. The two gate electrodes **74** included in the read transistor Mrd and the one gate electrode **68** included in the source follower transistor Msf are arranged in the second direction Dy so as to overlap the second semiconductor layer **65**. With this configuration, the source follower transistor Msf having a single-gate structure and the read transistor Mrd having a double-gate structure include the common second semiconductor layer **65**.
(71) Such a configuration can arrange the transistors and the wiring more efficiently than when forming each of the read transistor Mrd and the source follower transistor Msf from an individual semiconductor layer. In the present embodiment, the read transistor Mrd has a double-gate structure, so that a leakage current can be restrained from flowing toward the output signal line SL.
(72) As illustrated in FIGS. **5** and **6**, the photodiode **30** is provided in the region surrounded by the two of the reset control scan lines GLrst adjacent in the second direction Dy and the two of the output signal lines SL adjacent in the first direction Dx. The upper electrode **34** and the lower electrode **35** face each other with the photodiode **30** interposed therebetween in the third direction Dz. An overlapping electrode **37** is further provided so as to overlap the upper electrode **34**. Specifically, the photodiode **30** is disposed above the array substrate **2** provided with the various types of wiring and the various transistors with the lower electrode **35** interposed therebetween.
(73) The lower electrode **35** has a larger area than the photodiode **30** and the upper electrode **34** in the plan view. The lower electrode **35** is electrically coupled to the reset transistor Mrst and the

source follower transistor Msf through the contact hole H**2** at a portion overlapping neither the photodiode **30** nor the upper electrode **34**. The upper electrode **34** is provided so as to cover the photodiode **30**. A contact hole H**1** provided in an insulating film **27** is provided so as to overlap most of the region of the upper electrode **34**, and the insulating film **27** overlaps the upper electrode **34** only at the periphery of the upper electrode **34**.
(74) The overlapping electrode **37** is provided so as to cover the contact hole H**1** of the insulating film **27**, and is coupled to the upper electrode **34** in a region overlapping the contact hole H**1**. Coupling wiring **36** is coupled to the overlapping electrode **37** in the region thereof overlapping the contact hole H**1**. The above-described configuration electrically couples the photodiode **30** to reference potential supply wiring SLcom through the upper electrode **34**, the overlapping electrode **37**, and the coupling wiring **36**. The reference potential supply wiring SLcom is wiring for supplying the reference potential VCOM to the photodiode **30**, and is provided extending in the second direction Dy so as to overlap the output signal line SL.
(75) As illustrated in FIGS. **5** and **6**, the photodiode **30** and the lower electrode **35** are provided so as to overlap the various types of wiring and the various transistors (reset transistor Mrst, read transistor Mrd, and source follower transistor Msf). The photodiode **30** and the lower electrode **35** are provided so as to partially overlap the signal lines and the scan lines (power supply signal line SLsf, reset signal line SLrst, and read control scan line GLrd). A sensor region SA of the detection element **3** is defined by the lower electrode **35** coupled to the photodiode **30**. The optical sensitivity (sensor output) of the detection device **1** can be improved by increasing the area of the sensor region SA.
(76) The following describes a sectional configuration of the detection element **3**. FIG. **7** is a VII-VII′ sectional view of FIG. **5**. While FIG. **7** illustrates a sectional configuration of the reset transistor Mrst among the three transistors included in the detection element **3**, each of the source follower transistor Msf and the read transistor Mrd also has a sectional configuration similar to that of the reset transistor Mrst.
(77) As illustrated in FIG. **7**, the substrate **21** is an insulating substrate. A glass substrate of, for example, quartz or alkali-free glass is used as the substrate **21**. The substrate **21** has a first principal surface S**1**, and a second principal surface S**2** on the opposite side of the first principal surface S**1**. The various transistors including the reset transistor Mrst, the various types of wiring (scan lines and signal lines), and insulating films are provided on the first principal surface S**1** of the substrate **21** to form the array substrate **2**. The photodiodes **30** are arranged above the array substrate **2**, that is, on the first principal surface S**1** side of the substrate **21**.
(78) An undercoat film **22** is provided on the first principal surface S**1** of the substrate **21**. The undercoat film **22**, insulating films **23**, **24**, and **25**, the insulating film **27** and an insulating film **28** are inorganic insulating films, and are formed of, for example, silicon oxide (SiO.sub.2) or silicon nitride (SiN).
(79) The first semiconductor layer **61** is provided above the undercoat film **22**. For example, polysilicon is used as the first semiconductor layer **61**. The first semiconductor layer **61** is, however, not limited thereto, and may be formed of, for example, a microcrystalline oxide semiconductor, an amorphous oxide semiconductor, or low-temperature polycrystalline silicon (LTPS).
(80) The insulating film **23** is provided above the undercoat film **22** so as to cover the first semiconductor layer **61**. The gate electrodes **64** are provided above the insulating film **23**. The gate electrode **68** of the source follower transistor Msf is provided in the same layer as that of the gate electrodes **64**, and is also provided above the insulating film **23**. The reset control scan line GLrst and the read control scan line GLrd are also provided in the same layer as that of the gate electrodes **64**. The insulating film **24** is provided above the insulating film **23** so as to cover the gate electrodes **64**.
(81) As illustrated in FIG. **7**, the reset transistor Mrst has a top-gate structure in which the gate

electrodes **64** are provided above the first semiconductor layer **61**. However, in the detection device **1** of the present disclosure, the reset transistor Mrst may have a bottom-gate structure in which the gate electrodes **64** are provided below the first semiconductor layer **61**, or a dual-gate structure in which the gate electrodes **64** are provided above and below the first semiconductor layer **61**.

(82) The insulating films **24** and **25** are provided above the insulating film **23** so as to cover the gate electrodes **64**. The source electrode **62** and the drain electrode **63** are provided above the insulating film **25**. The insulating films **24** and **25** serve as a first interlayer insulating film that insulates the gate electrodes **64** from the source and the drain electrodes **62** and **63**. The source and the drain electrodes **62** and **63** are each coupled to the first semiconductor layer **61** through a contact hole passing through the insulating films **23**, **24**, and **25**. The source and the drain electrodes **62** and **63** are formed of, for example, a multilayered film of Ti—Al—Ti layers or Ti—Al layers that has a multilayered structure of titanium and aluminum.

(83) The various signal lines (output signal line SL, power supply signal line SLsf, and reset signal line SLrst) and the coupling wiring SLcn are provided in the same layer as that of the source and the drain electrodes **62** and **63**. The coupling wiring SLcn of the detection element **3** is coupled to the gate electrode **68** of the source follower transistor Msf through a contact hole passing through the insulating films **24** and **25**.

(84) As illustrated in FIG. **7**, an insulating film **26** is provided above the insulating film **25** so as to cover the various transistors, including the reset transistor Mrst, for example. The insulating film **26** is formed of an organic material such as a photosensitive acrylic. The insulating film **26** is thicker than the insulating film **25**. The insulating film **26** has a better step covering property than that of inorganic insulating materials, and can planarize steps formed by the various transistors and the various types of wiring.

(85) The following describes a sectional configuration of the photodiode **30**. The photodiode **30** is provided above the insulating film **26**. Specifically, the lower electrode **35** is provided above the insulating film **26**, and is electrically coupled to the coupling wiring SLcn through the contact hole H**2**. The photodiode **30** is coupled to the lower electrode **35**. The lower electrode **35** can employ, for example, a multilayered structure of titanium (Ti) and titanium nitride (TiN). Since the lower electrode **35** is provided between the substrate **21** and the photodiode **30**, the lower electrode **35** serves as a light-blocking layer, and can restrain light from entering the photodiode **30** from the second principal surface S**2** side of the substrate **21**.

(86) The photodiode **30** includes a semiconductor layer having a photovoltaic effect. Specifically, the semiconductor layer of the photodiode **30** includes an i-type semiconductor layer **31**, a p-type semiconductor layer **32**, and the n-type semiconductor layer **33**. The i-type semiconductor layer **31**, the p-type semiconductor layer **32**, and the n-type semiconductor layer **33** are formed of, for example, amorphous silicon (a-Si). The material of the semiconductor layers is not limited thereto, and may be, for example, polysilicon or microcrystalline silicon.

(87) The a-Si of the p-type semiconductor layer **32** is doped with impurities to form a p+ region. The a-Si of the n-type semiconductor layer **33** is doped with impurities to form an n+ region. The i-type semiconductor layer **31** is, for example, a non-doped intrinsic semiconductor, and has lower electric conductivity than that of the p-type semiconductor layer **32** and the n-type semiconductor layer **33**.

(88) The i-type semiconductor layer **31** is provided between the n-type semiconductor layer **33** and the p-type semiconductor layer **32** in the direction orthogonal to the surface of the substrate **21** (in the third direction Dz). In the present embodiment, the n-type semiconductor layer **33**, the i-type semiconductor layer **31**, and the p-type semiconductor layer **32** are stacked in this order above the lower electrode **35**. In the present embodiment, the upper electrode **34** serves as the anode of the photodiode **30**, and the lower electrode **35** serves as the cathode of the photodiode **30**. The order of stacking of the n-type semiconductor layer **33**, the i-type semiconductor layer **31**, and the p-type semiconductor layer **32** may be reversed.

(89) The n-type semiconductor layer **33** of the photodiode **30** of the detection element **3** is electrically coupled to the reset transistor Mrst and the source follower transistor Msf through the lower electrode **35** and the coupling wiring SLcn.
(90) The upper electrode **34** is provided above the p-type semiconductor layer **32**. The upper electrode **34** is formed of, for example, a light-transmitting conductive material such as indium tin oxide (ITO). The insulating film **27** is provided above the insulating film **26** so as to cover the photodiode **30** and the upper electrode **34**. The insulating film **27** is provided with the contact hole H**1** (opening) in a region overlapping the upper electrode **34**.
(91) The overlapping electrode **37** is provided above the upper electrode **34**, and is electrically coupled to the upper electrode **34** at the bottom of the contact hole H**1**. The overlapping electrode **37** is formed of a light-transmitting conductive material such as ITO that is the same as the material of the upper electrode **34**. The overlapping electrode **37** is provided along the inner wall of the contact hole H**1** and the upper surface of the insulating film **27**, and covers the boundary portion between the upper electrode **34** and the insulating film **27**. The coupling wiring **36** is coupled to the overlapping electrode **37** in the region thereof overlapping the contact hole H**1**. The above-described configuration electrically couples the photodiode **30** to the reference potential supply wiring SLcom through the upper electrode **34**, the overlapping electrode **37**, and the coupling wiring **36**. The p-type semiconductor layer **32** is supplied with the reference potential VCOM (refer to FIG. **4**) through the coupling wiring **36**, the overlapping electrode **37**, and the upper electrode **34**.
(92) As illustrated in FIG. **6**, the overlapping electrode **37** is provided so as to cover the entire region of the contact hole H**1**. In other words, the inner wall of contact hole H**1** is located inside the outer periphery of the overlapping electrode **37**. In the present embodiment, the overlapping electrode **37** serves as a protective film for the photodiode **30**, and can restrain water or the like from entering the photodiode **30** side. When layers (for example, the coupling wiring **36** and the reference potential supply wiring SLcom) above the photodiode **30** are patterned in the manufacturing process of a detection device **1**B, the overlapping electrode **37** serves as the protective film to restrain the photodiode **30** from being damaged. For example, since the overlapping electrode **37** is provided so as to cover the interface between the upper electrode **34** and the insulating film **27**, the etchant can be restrained from entering the photodiode **30** side through the interface. However, the overlapping electrode **37** of the photodiode **30** may not be formed. In that case, the structure is such that the coupling wiring **36** is in direct contact with the upper electrode **34**. By not forming the overlapping electrode **37** over the photodiode **30**, the photodiode **30** can be improved in detection accuracy as compared with the structure in which the overlapping electrode **37** overlaps the upper electrode **34**.
(93) As illustrated in FIG. **7**, the insulating film **28** is provided above the insulating film **27** so as to cover the upper electrode **34**, the overlapping electrode **37**, and the coupling wiring **36**. The insulating film **28** is provided as a protective layer for restraining water from entering the photodiode **30**. In addition, an insulating film **29** is provided above the insulating film **28**. The insulating film **29** is a hard coat film formed of an organic material. The insulating film **29** planarizes steps formed on a surface of the insulating film **28** by the photodiode **30** and the coupling wiring **36**.
(94) The cover member **122** is provided so as to face the array substrate **2**. That is, the cover member **122** is provided so as to cover the various transistors and the photodiode **30**. In more detail, the adhesive layer **125** bonds the insulating film **29** of the array substrate **2** to the cover member **122**. The adhesive layer **125** is, for example, a light-transmitting optical clear adhesive (OCA) sheet.
(95) As described above, in the present embodiment, the insulating film **26** is provided so as to cover the transistors such as the reset transistor Mrst, and the lower electrode **35**, the photodiode **30**, and the upper electrode **34** are stacked in this order above the insulating film **26**. Since the

lower electrode **35**, the photodiode **30**, and the upper electrode **34** are provided in layers different from those of the transistors, the signal lines, and the scan lines, the degree of freedom of arrangement of the photodiode **30** and the lower electrode **35** can be improved, as described above.
(96) The following describes a terminal portion T that electrically couples the array substrate **2** to the wiring substrate **110**. FIG. **8** is a plan view schematically illustrating a configuration of the terminal portion of the detection device according to the first embodiment. As illustrated in FIG. **8**, the terminal portion T includes a plurality of first terminals **81** and a plurality of second terminals **82**. The first terminals **81** are arranged in the first direction Dx in the peripheral region GA of the substrate **21**. Each of the first terminals **81** extends in the second direction Dy, and an end thereof in the second direction Dy is coupled to signal line coupling wiring **85** through a first contact portion CN**1**. Each wire of the signal line coupling wiring **85** extends in the second direction Dy, and is electrically coupled to the output signal line SL in the detection region AA through the signal line selection circuit **16** (refer to FIG. **2**). The first terminals **81** output the detection signals Vdet from the photodiodes **30** in the detection region AA to external circuitry. The signal line coupling wiring **85** need not be coupled to the end of the first terminal **81**. The location where the first terminal **81** is coupled to the signal line coupling wiring **85** need not be the end of the first terminal **81**, and may be, for example, the center of the first terminal **81**. That is, the location of the first contact portion CN**1** is not limited to the end of the first terminal **81**.
(97) The second terminals **82** are arranged in the first direction Dx with the first terminals **81** interposed between two groups of the second terminals **82**. That is, the second terminals **82**, the first terminals **81**, and the second terminals **82** are arranged in this order in the first direction Dx. Ends in the second direction Dy of the second terminals **82** are coupled to drive circuit coupling wiring **86** through second contact portions CN**2**. The drive circuit coupling wiring **86** is drawn in the peripheral region GA, and electrically coupled to the scan line drive circuit **15**. In the same manner as the locations of formation of the first contact portions CN**1** described above, the locations where the second contact portions CN**2** are formed are not limited to the ends of the second terminals **82**.
(98) For example, the second terminals **82** arranged on the left side of FIG. **8** are electrically coupled to the first scan line drive circuit **15**A (refer to FIG. **2**), and supply various scan signals and control signals to the first scan line drive circuit **15**A. The second terminals **82** arranged on the right side of FIG. **8** are electrically coupled to the second scan line drive circuit **15**B (refer to FIG. **2**), and supply various scan signals and control signals to the second scan line drive circuit **15**B. The second terminals **82** include terminals that are directly or indirectly coupled to the signal lines in the detection region AA to supply thereto the reset potential Vrst and the power supply potential VDD.
(99) The first and the second terminals **81** and **82** are each electrically coupled to a wiring substrate terminal **111** (refer to FIG. **12**) of the wiring substrate **110** through an anisotropic conductive film **112** (ACF).
(100) A dummy terminal region **83** is provided between the second terminals **82** adjacent in the first direction Dx. The dummy terminal region **83** is a region provided with no metal layer, and is a region not coupled to the wiring substrate terminal **111** (refer to FIG. **12**) of the wiring substrate **110**.
(101) The inclination of the first and the second terminals **81** and **82** with respect to the second direction Dy is smaller in the center of the first direction Dx, and increases toward the outer periphery sides in the first direction Dx of the substrate **21**. However, the first and the second terminals **81** and **82** are not limited to this arrangement, and may be arranged in parallel. FIG. **8** is merely a schematic illustration, and the number, the shape, the arrangement pitch, and the like of the first and the second terminals **81** and **82** may be changed as appropriate.
(102) In the present embodiment, what is called a film-on-glass (FOG) mounting is used in which the wiring substrate **110** is mounted on the first and the second terminals **81** and **82** using the

anisotropic conductive film **112**. However, the present embodiment is not limited thereto, and can also be applied to what is called a chip-on-glass (COG) mounting in which an integrated circuit (IC) is mounted on the first and the second terminals **81** and **82**.
(103) The following describes a detailed configuration of the first terminal **81**. The following description of the first terminal **81** is also applicable to each of the second terminals **82**. FIG. **9** is a plan view illustrating a magnified view of the first terminal. FIG. **10** is a X-X′ sectional view of FIG. **9**.
(104) As illustrated in FIG. **9**, the first terminal **81** includes a first metal layer ML**1**, a second metal layer ML**2**, a third metal layer ML**3**, and a first light-transmitting conductive layer CL**1**. The first metal layer ML**1**, the second metal layer ML**2**, the third metal layer ML**3**, and the first light-transmitting conductive layer CL**1** are rectangular in the plan view, and are provided so as to overlap one another.
(105) The first metal layer ML**1** is provided in the same layer as that of the signal line coupling wiring **85**, and is coupled to the signal line coupling wiring **85** at an end in the second direction Dy of the first metal layer ML**1**. The first metal layer ML**1** is electrically coupled to the second metal layer ML**2** through the first contact portions CN**1**. The third metal layer ML**3** and the first light-transmitting conductive layer CL**1** are stacked in this order above the second metal layer ML**2**.
(106) As illustrated in FIG. **10**, the undercoat film **22** and the insulating films **23**, **24**, and **25** of array substrate **2** are continuously formed from the detection region AA to the peripheral region GA in which the first terminal **81** is provided. The first metal layer ML**1** is provided above the insulating film **23**. That is, the first metal layer ML**1** is the same layer as that of the gate electrodes **64** (refer to FIG. **7**) and the scan lines (reset control scan line GLrst and read control scan line GLrd) provided in the detection region AA, and is formed of the same material as that of the gate electrodes **64** and the scan lines.
(107) The second metal layer ML**2** is stacked above the first metal layer ML**1** with the insulating films **24** and **25** (first interlayer insulating film) interposed therebetween. That is, the second metal layer ML**2** is the same layer as that of the various signal lines such as the output signal line SL and the reset signal line SLrst (refer to FIG. **7**) provided in the detection region AA, and is formed of the same material as that of the various signal lines.
(108) The third metal layer ML**3** is stacked above the second metal layer ML**2** so as to be in contact therewith. The third metal layer ML**3** is the same layer as that of the lower electrode **35** (refer to FIG. **7**), and is formed of the same material as that of the lower electrode **35**. The insulating film **26** illustrated in FIG. **7** is not provided in a region of the peripheral region GA overlapping at least the terminal portion T, and the third metal layer ML**3** directly contacts the top of the second metal layer ML**2** without the insulating film **26** interposed therebetween.
(109) The first light-transmitting conductive layer CL**1** is stacked above the third metal layer ML**3** so as to be in contact therewith. The first light-transmitting conductive layer CL**1** is the same layer as that of the overlapping electrode **37** (refer to FIG. **7**), and is formed of the same material as that of the overlapping electrode **37**, such as ITO.
(110) The insulating film **28** is provided above the insulating film **25** so as to cover the first light-transmitting conductive layer CL**1** of the first terminal **81**. The insulating film **28** is provided with an opening OP to expose the first light-transmitting conductive layer CL**1** in a region overlapping each of the first terminals **81**. In other words, the insulating film **28** is provided so as to cover the periphery of the first light-transmitting conductive layer CL**1**, and the opening OP is formed inside the periphery. The first light-transmitting conductive layer CL**1** is provided at least in a region overlapping the opening OP. The insulating film **28** is provided between the first terminals **81** adjacent in the first direction Dx to insulate between the first terminals **81**. The first light-transmitting conductive layer CL**1** is not limited to being the same layer as that of the overlapping electrode **37** (refer to FIG. **7**), and may be the same layer as that of the upper electrode **34**. In this case, the insulating film **27** may be provided instead of the insulating film **28**.

(111) A width W**1** in the first direction Dx of the first metal layer ML**1** is equal to a width W**2** in the first direction Dx of the opening OP. The width in the first direction Dx of the second metal layer ML**2** is larger than each of the widths W**1** and W**2**. The width in the first direction Dx of the third metal layer ML**3** is larger than the width in the first direction Dx of the second metal layer ML**2**. The width in the first direction Dx of the first light-transmitting conductive layer CL**1** is smaller than the width in the first direction Dx of the third metal layer ML**3**.
(112) FIG. **11** is an XI-XI′ sectional view of FIG. **9**. As illustrated in FIG. **11**, the first metal layer ML**1** is formed to have a width in the second direction Dy larger than the width in the second direction Dy of the opening OP. The first contact portions CN**1** are provided on the second direction Dy side (detection region AA side) of the opening OP, and pass through the insulating films **24** and **25**. The first metal layer ML**1** is coupled to the second metal layer ML**2** through the first contact portions CN**1**. In FIG. **11**, three of the first contact portions CN**1** are arranged in the second direction Dy.
(113) FIG. **12** is a sectional view illustrating a structure example of bonding between the array substrate and the wiring substrate according to the first embodiment. As illustrated in FIG. **12**, the wiring substrate terminal **111** of the wiring substrate **110** faces the first terminal **81** with the anisotropic conductive film **112** interposed therebetween. The anisotropic conductive film **112** includes a resin layer **113** and a number of conductive particles **114** dispersed in the resin layer **113**, and is provided above the insulating film **28** so as to cover the first terminal **81**. The anisotropic conductive film **112** is in direct contact with a side surface of the insulating film **28** forming the opening OP and with the first light-transmitting conductive layer CL**1** overlapping the opening OP. Pressure-bonding the wiring substrate **110** electrically couples the wiring substrate terminal **111** to the first light-transmitting conductive layer CL**1** through the conductive particles **114**.
(114) Thus, the first, the second, and the third metal layers ML**1**, ML**2**, and ML**3**, and the first light-transmitting conductive layer CL**1** of the first terminals **81** are formed of the same layers as the metal layers and the light-transmitting conductive layers that constitute the photodiodes **30** and the transistors in the detection region AA. This configuration allows the first terminals **81** to be formed in the same process as the manufacturing process of the detection region AA. Therefore, the manufacturing process of the detection device **1** can be restrained from expanding, and the manufacturing cost can be reduced.
(115) The first light-transmitting conductive layer CL**1** is provided so as to cover the second and the third metal layers ML**2** and ML**3**. The first light-transmitting conductive layer CL**1** serves as a protective layer, and can reduce corrosion or the like of the second and the third metal layers ML**2** and ML**3**. An effective coupling region between the wiring substrate **110** and the first terminal **81** is defined by the area of the opening OP provided in the insulating film **28**. In the present embodiment, since the insulating film **28** is provided above the first light-transmitting conductive layer CL**1**, the first light-transmitting conductive layer CL**1** can be formed larger with a large contact area ensured between the first light-transmitting conductive layer CL**1** and the anisotropic conductive film **112** in the opening OP, while avoiding a short circuit of the first light-transmitting conductive layer CL**1** between the adjacent first terminals **81** or between the adjacent second terminals **82**. Furthermore, as will be described later, the opening OP can be formed large enough to occupy most of the area of the first metal layer ML**1**. This configuration can ensure the coupling reliability between the wiring substrate **110** and the first terminal **81**.
(116) Since the first metal layer ML**1** is provided so as to overlap the opening OP and is formed to have the width W**1** same as the width W**2**, the flatness of the first light-transmitting conductive layer CL**1** can be improved in the region overlapping the opening OP. By observing the state of a surface (surface on the substrate **21** side) of the first metal layer ML**1**, the coupling state between the first terminal **81** and the wiring substrate **110** (pressure-bonding state of the anisotropic conductive film **112**) can be checked well.
(117) The following describes a configuration of the dummy terminal region **83**. FIG. **13** is a XIII-

XIII′ sectional view of FIG. **8**. As illustrated in FIG. **13**, the insulating film **27** (second interlayer insulating film) is provided between the insulating films **24** and **25** (first interlayer insulating film) and the insulating film **28** in the dummy terminal regions **83** between the adjacent second terminals **82** and between the adjacent first and second terminals **81** and **82**.

(118) This configuration can reduce the difference between the height position of the first light-transmitting conductive layers CL**1** of the first and the second terminals **81** and **82** and the height position of the insulating film **28** in the dummy terminal region **83** as compared with a case where the insulating film **27** is not provided. As a result, when the wiring substrate **110** is mounted, the difference between the force applied to the first and the second terminals **81** and **82** and the force applied to the dummy terminal region **83** can be reduced to improve the stability of the mounting. When compared with a case where a metal layer is provided instead of the insulating film **27** (second interlayer insulating film) as a height adjusting structure for the dummy terminal region **83**, an unintended electric field can be restrained from being generated between the metal layer of the dummy terminal region **83** and each of the metal layers of the second terminal **82**. As a result, a possible occurrence of electric field corrosion of the second terminal **82** under a high-temperature high-humidity environment can be restrained.

(119) The "height position" refers to the distance in the third direction Dz between the first principal surface S**1** of the substrate **21** (refer to FIG. **7**) and the surface of the first light-transmitting conductive layer CL**1** of each of the first and the second terminals **81** and **82**. Alternatively, the "height position" refers to the distance in the third direction Dz between the first principal surface S**1** of the substrate **21** (refer to FIG. **7**) and the surface of the insulating film **28** in the dummy terminal region **83**.

(120) In FIG. **13**, the width of the dummy terminal region **83** (right side the dummy terminal region **83**) between the adjacent second terminals **82** is larger than the width of the dummy terminal region **83** (left side dummy terminal region **83**) between the adjacent first and second terminals **81** and **82**. The dummy terminal region **83** may be formed to have the width of one of the first terminals **81** (or second terminals **82**) or to have a width of two or more of the first terminals **81** (or second terminals **82**).

Second Embodiment

(121) FIG. **14** is a sectional view schematically illustrating a first terminal according to a second embodiment. In the following description, the same components as those described in the embodiment described above are denoted by the same reference numerals, and the description thereof will not be repeated.

(122) The following describes a detection device **1**A of the second embodiment having a configuration in which the order of stacking the first light-transmitting conductive layer CL**1** and the insulating film **28** of a first terminal **81**A differs from that of the first embodiment described above. Specifically, as illustrated in FIG. **14**, the insulating film **28** is provided above the insulating film **25** so as to cover the third metal layer ML**3**. The insulating film **28** is provided with the opening OP in a region thereof overlapping the third metal layer ML**3**.

(123) The first light-transmitting conductive layer CL**1** is provided so as to cover the insulating film **28** and the opening OP, and is stacked above the third metal layer ML**3** so as to be in contact therewith in a region overlapping the opening OP. The first light-transmitting conductive layer CL**1** is provided so as to cover the edge of the insulating film **28** around the opening OP.

(124) In the present embodiment, since the first light-transmitting conductive layer CL**1** is provided so as to cover the boundary portion between the insulating film **28** and the third metal layer ML**3**, water or the like can be restrained from entering the third metal layer ML**3** side and the second metal layer ML**2** side from the boundary portion. Even in this case, the first light-transmitting conductive layer CL**1** provided above the insulating film **28** outside the opening OP is electrically coupled to the conductive particles **114** (refer to FIG. **12**). The first light-transmitting conductive layer CL**1** provided in a region outside the opening OP can also be used as an auxiliary coupling

region, thus being capable of increasing the contact area between the anisotropic conductive film **112** and the first light-transmitting conductive layer CL**1**. This configuration can restrain increase in coupling resistance between the first terminal **81** and the wiring substrate **110** also in the second embodiment.

Third Embodiment

(125) FIG. **15** is a sectional view schematically illustrating a first terminal according to a third embodiment. The following describes a detection device **1**B of the third embodiment having a configuration in which a first terminal **81**B includes a second light-transmitting conductive layer CL**2** as compared with the first and the second embodiments described above.

(126) Specifically, as illustrated in FIG. **15**, the second light-transmitting conductive layer CL**2** is stacked above the first light-transmitting conductive layer CL**1** so as to be in contact therewith. The insulating films **28** is provided above the insulating film **25** so as to cover the second light-transmitting conductive layer CL**2**. The opening OP is provided in a region of the insulating film **28** overlapping the second light-transmitting conductive layer CL**2**. That is, the first and the second light-transmitting conductive layers CL**1** and CL**2** are provided at least in a region overlapping the opening OP. The width in the first direction Dx of the second light-transmitting conductive layer CL**2** is equal to the width in the first direction Dx of the first light-transmitting conductive layer CL**1**. The width in the first direction Dx of the first and the second light-transmitting conductive layers CL**1** and CL**2** is larger than each of the widths W**1** and W**2**. The width of the second light-transmitting conductive layer CL**2** is not limited to this width, and may differ from the width of the first light-transmitting conductive layer CL**1**.

(127) The first light-transmitting conductive layer CL**1** is the same layer as that of the upper electrode **34** (refer to FIG. **7**), and is formed of the same material as that of the upper electrode **34**. The second light-transmitting conductive layer CL**2** is the same layer as that of the overlapping electrode **37** (refer to FIG. **7**), and is formed of the same material as that of the overlapping electrode **37**.

(128) In the present embodiment, since the second light-transmitting conductive layer CL**2** is stacked above the first light-transmitting conductive layer CL**1**, the contact resistance of the first terminal **81**B can be reduced from that of each of the first and the second embodiments described above. In the present embodiment, the first and the second light-transmitting conductive layers CL**1** and CL**2** can protect the second and the third metal layers ML**2** and ML**3**.

(129) The present embodiment can be combined with the second embodiment described above. That is, in FIG. **14**, the second light-transmitting conductive layer CL**2** may be stacked above the first light-transmitting conductive layer CL**1**. In other words, the first and the second light-transmitting conductive layers CL**1** and CL**2** may be provided so as to cover the opening OP and the edge of the insulating film **28** around the opening OP. Alternatively, the first light-transmitting conductive layer CL**1**, the insulating film **28**, and the second light-transmitting conductive layer CL**2** may be stacked in this order.

Fourth Embodiment

(130) FIG. **16** is a plan view schematically illustrating a configuration of a terminal portion of a detection device according to a fourth embodiment. The following describes a detection device **1**C of the third embodiment having a configuration in which a terminal portion TA includes outer edge wiring **87** and **88** as compared with the first to the third embodiments describe above.

(131) As illustrated in FIG. **16**, the outer edge wiring **87** is electrically coupled to each of the first terminals **81**, and extends in the second direction Dy between the first terminals **81** and an outer periphery **21***e* of the substrate **21**. The outer edge wiring **88** is electrically coupled to each of the second terminals **82**, and extends in the second direction Dy between the second terminals **82** and the outer periphery **21***e* of the substrate **21**. Ends of the outer edge wiring **87** and **88** overlap the outer periphery **21***e* of the substrate **21**.

(132) Wires of the outer edge wiring **87** and **88** are electrically coupled together in a region outside

the outer periphery **21***e* before the substrate **21** is cut to an external shape, and short-circuit the first and the second terminals **81** and **82**. The wires of the outer edge wiring **87** and **88** are what are called short rings provided as a countermeasure for electrostatic discharge (ESD). In the substrate **21** after being cut to the external shape, the ends on the outer periphery **21***e* side of the outer edge wiring **87** and **88** are coupled to neither the other adjacent wires of the outer edge wiring **87** and **88** nor the other first and second terminals **81** and **82**.

(133) In FIG. **16**, the outer edge wiring **87** and **88** are coupled to the first and the second terminals **81** and **82**, respectively. However, the present embodiment is not limited thereto. The outer edge wiring **88** only needs to be coupled to at least more than one of the second terminals **82**, and the outer edge wiring **87** need not be coupled to more than one of the first terminals **81**.

(134) FIG. **17** is a plan view illustrating a magnified view of the first terminal according to the fourth embodiment. FIG. **18** is an XVIII-XVIII′ sectional view of FIG. **17**. The following describes configurations of the first terminal **81** and the outer edge wiring **87** with reference to FIGS. **17** and **18**, but the description of the first terminal **81** and the outer edge wiring **87** is also applicable to the second terminal **82** and the outer edge wiring **88**.

(135) As illustrated in FIG. **17**, the outer edge wiring **87** is arranged adjacent in the second direction Dy to the signal line coupling wiring **85** and the opening OP, and overlaps neither the first metal layer ML**1** (signal line coupling wiring **85**) nor the opening OP. An end side in the second direction Dy of the outer edge wiring **87** is provided so as to overlap a portion of the second metal layer ML**2**, and is electrically coupled to the second metal layer ML**2** through third contact portions CN**3**. The outer edge wiring **87** is formed of, for example, a semiconductor layer PS, and is formed of a material different from those of the first to the third metal layers ML**1** to ML**3** and the first light-transmitting conductive layer CL**1**. The outer edge wiring **87** is formed of a material having higher resistance than the first to the third metal layers ML**1** to ML**3**.

(136) As illustrated in FIG. **18**, the outer edge wiring **87** is provided above the undercoat film **22**. That is, the outer edge wiring **87** is the same layer as the first semiconductor layer **61** (refer to FIG. **7**), and is formed of the same material as that of the first semiconductor layer **61** (refer to FIG. **7**).

(137) The first metal layer ML**1** is provided in a region overlapping the opening OP, and is not provided in a region between the opening OP and the outer periphery **21***e* of the substrate **21** in the second direction Dy. The outer edge wiring **87** is provided in a region between ends in the second direction Dy of the first metal layer ML**1** and the opening OP and the outer periphery **21***e* of the substrate **21**.

(138) The second metal layer ML**2**, the third metal layer ML**3**, and the first light-transmitting conductive layer CL**1** are provided so as to extend from the region overlapping the opening OP to the outer periphery **21***e* side of the substrate **21**. The third contact portions CN**3** are provided through the insulating films **23**, **24**, and **25**. The second metal layer ML**2** is coupled to the outer edge wiring **87** through the third contact portions CN**3**

(139) FIG. **19** is an explanatory view for explaining an arrangement relation among the first terminals, the second terminals, the outer edge wiring, and the insulating films. FIG. **20** is a XX-XX′ sectional view of FIG. **19**. For ease of viewing the drawings, FIGS. **19** and **20** schematically illustrate the outer edge wiring **87** and **88** coupled to two of the first terminals **81** and two of the second terminals **82**.

(140) As illustrated in FIG. **19**, the insulating film **23** and the insulating films **24** and **25** (first interlayer insulating film) are provided in regions overlapping the first and the second terminals **81** and **82**, and also provided in a region between the first and the second terminals **81** and **82** and the outer periphery **21***e* of the substrate **21**.

(141) In the region between the first and the second terminals **81** and **82** and the outer periphery **21***e* of the substrate **21**, the insulating film **23** and the insulating films **24** and **25** (first interlayer insulating film) extend in the second direction Dy so as to cover each wire of the outer edge wiring **87** and **88**. In regions between the wires of the outer edge wiring **87** and **88** adjacent in the first

direction Dx, the insulating film **23** and the insulating films **24** and **25** (first interlayer insulating film) are removed to form cutouts NT.
(142) The insulating film **28** is provided so as to cover the first and the second terminals **81** and **82**, and also covers portion of the cutouts NT. More specifically, the insulating film **28** is provided so as to cover stepped portions at locations of the cutouts NT closer to the first and the second terminals **81** and **82**.
(143) As illustrated in FIG. **20**, the outer edge wiring **87** coupled to the first terminals **81** and the insulating film **23** and the insulating films **24** and **25** (first interlayer insulating film) that cover the outer edge wiring **87** are stacked to form a plurality of convex portions **89**. The outer edge wiring **88** coupled to the second terminals **82** and the insulating film **23** and the insulating films **24** and **25** (first interlayer insulating film) that cover the outer edge wiring **88** are stacked to form a plurality of convex portions **90**. The convex portions **89** and **90** are arranged in the first direction Dx. In the cutouts NT between the adjacent convex portions **89** and **90**, the insulating film **23** and the insulating films **24** and **25** (first interlayer insulating film) are not provided, and the undercoat film **22** is located on the surface.
(144) Since the convex portions **89** and **90** are provided, the resin layer **113** (refer to FIG. **12**) of the anisotropic conductive film **112** is provided so as to cover the convex portions **89** and **90** when the wiring substrate **110** has been coupled to the first and the second terminals **81** and **82**. This configuration increases the contact area between the resin layer **113** of the anisotropic conductive film **112** and the convex portions **89** and **90** in the region between the first and the second terminals **81** and **82** and the outer periphery **21***e* of the substrate **21** as compared with a configuration in which the insulating film **23** and the insulating films **24** and **25** (first interlayer insulating film) are flatly provided. As a result, in the present embodiment, the strength of coupling of the first and the second terminals **81** and **82** to the wiring substrate **110** can be increased.
(145) FIG. **21** is an explanatory view for explaining an arrangement relation among the first terminals, the second terminals, the outer edge wiring, and the insulating films according to a fourth modification of the fourth embodiment. FIG. **22** is a XXII-XXII′ sectional view of FIG. **21**. The following describes a detection device **1**D according to the fourth modification of the fourth embodiment having a configuration in which the first terminal **81** is not provided with the outer edge wiring **87**.
(146) As illustrated in FIG. **21**, the insulating film **23** and the insulating films **24** and **25** (first interlayer insulating film) are provided between the first terminals **81** and the outer periphery **21***e* of the substrate **21**. The insulating film **23** and the insulating films **24** and **25** (first interlayer insulating film) are provided with the cutouts NT in the same manner as in the fourth embodiment described above. As a result, convex portions **89**A (first convex portions) are provided in a region between the first terminals **81** and the outer periphery **21***e* of the substrate **21**, and extend in the second direction Dy. The convex portions **90** (second convex portions) are provided in regions between the second terminals **82** and the outer periphery **21***e* of the substrate **21**, and extend in the second direction Dy.
(147) As illustrated in FIG. **22**, the convex portions **89**A (first convex portions) do not include the outer edge wiring **87**, and are formed by stacking the insulating film **23** and the insulating films **24** and **25** (first interlayer insulating film). The convex portions **90** (second convex portions) are formed by stacking the outer edge wiring **88**, the insulating film **23** covering the outer edge wiring **88**, and the insulating films **24** and **25**. The convex portions **89**A (first convex portions) and the convex portions **90** (second convex portions) are arranged in the first direction Dx.
(148) In the fourth modification, the convex portions **89**A (first convex portions) are provided correspondingly to the respective first terminals **81** even if the outer edge wiring **87** is not coupled to the first terminals **81**. As a result, also in the fourth modification, the strength of coupling of the first and the second terminals **81** and **82** to the wiring substrate **110** can be increased in the same manner as in the fourth embodiment in which the outer edge wiring **87** is coupled to the first

terminals **81**.
(149) FIG. **23** is an explanatory view for explaining an arrangement relation among the first terminals, the second terminals, the outer edge wiring, and the insulating films according to a fifth modification of the fourth embodiment. FIG. **24** is a XXIV-XXIV′ sectional view of FIG. **23**. The following describes a detection device **1**E according to the fifth modification of the fourth embodiment having a configuration in which one continuous convex portion **91** is provided in the region between the first and the second terminals **81** and **82** and the outer periphery **21***e* of the substrate **21**.
(150) As illustrated in FIGS. **23** and **24**, the convex portion **91** is formed by stacking a plurality of wires of the outer edge wiring **88**, the insulating film **23** covering the wires of the outer edge wiring **88**, and the insulating films **24** and **25**. The convex portion **91** extends in the first direction Dx adjacent to the first and the second terminals **81** and **82**. The insulating film **23** and the insulating films **24** and **25** are provided with the cutouts NT adjacent in the first direction Dx to the convex portion **91**.
(151) As illustrated in FIG. **23**, the insulating film **28** covers the first and the second terminals **81** and **82**, and is provided so as not to overlap the cutouts NT.
(152) In the fifth modification as compared with the fourth embodiment and the fourth modification described above, no cutouts NT are provided between the wires of the outer edge wiring **88**. Therefore, insulation between the adjacent wires of the outer edge wiring **88** can be ensured. In the fifth modification, the outer edge wiring **87** is not coupled to the first terminals **81**, and the convex portion **91** is provided in the region between the first terminals **81** and the outer periphery **21***e* of the substrate **21**. However, the present modification is not limited to this configuration. The outer edge wiring **87** may be coupled to the first terminals **81**. In that case, the one convex portion **91** is provided so as to cover the wires of the outer edge wiring **87** and the wires of the outer edge wiring **88**.
(153) The configuration of each of the fourth embodiment and the fourth and the fifth modifications can be combined with any one of the first to the third embodiments described above.
(154) While the preferred embodiments of the present invention have been described above, the present invention is not limited to the embodiments described above. The content disclosed in the embodiments is merely an example, and can be variously modified within the scope not departing from the gist of the present invention. Any modifications appropriately made within the scope not departing from the gist of the present invention also naturally belong to the technical scope of the present invention.

Claims

1. A detection device comprising: a substrate; a plurality of photodiodes that are arranged in a detection region of the substrate; a plurality of first terminals that are arranged in a first direction in a peripheral region outside the detection region of the substrate; an insulating film that covers the first terminals; and an anisotropic conductive film that is located above the insulating film, and covers the first terminals, wherein each of the first terminals comprises, between the substrate and the insulating film: a first metal layer; a second metal layer that is stacked above the first metal layer with a first interlayer insulating film interposed between the first metal layer and the second metal layer; a third metal layer that is stacked above the second metal layer so as to be in contact with the second metal layer; and a first light-transmitting conductive layer that is stacked above the third metal layer so as to be in contact with the third metal layer, the insulating film has an opening that exposes the first light-transmitting conductive layer in a region overlapping each of the first terminals, and the anisotropic conductive film is in direct contact with a side surface of the insulating film forming the opening and with the first light-transmitting conductive layer overlapping the opening.

2. The detection device according to claim 1, wherein the insulating film is provided so as to cover a periphery of the first light-transmitting conductive layer, and the opening is formed inside the periphery.
3. The detection device according to claim 1, further comprising: a plurality of scan lines that are provided in the detection region; and a plurality of output signal lines that are provided in the detection region, and intersect the scan lines, wherein the first metal layer is the same layer as that of the scan lines, and the second metal layer is the same layer as that of the output signal lines.
4. The detection device according to claim 3, further comprising a lower electrode and an upper electrode that are provided in the detection region, wherein the lower electrode, each of the photodiodes, and the upper electrode are stacked in this order, and the third metal layer is the same layer as that of the lower electrode.
5. The detection device according to claim 4, further comprising an overlapping electrode provided so as to overlap the upper electrode, wherein the first light-transmitting conductive layer is the same layer as that of the upper electrode or the overlapping electrode.
6. The detection device according to claim 4, further comprising, in the detection region, an inorganic insulating film that covers the upper electrode of the photodiode, and a coupling wiring coupled to the upper electrode through a contact hole formed in the inorganic insulating film, wherein the inorganic insulating film is formed between the substrate and the insulating film, and the first light-transmitting conductive layer is an electrode formed between the insulating film and the inorganic insulating film.
7. The detection device according to claim 5, further comprising a second light-transmitting conductive layer that is stacked on the first light-transmitting conductive layer, wherein the first light-transmitting conductive layer is the same layer as that of the upper electrode, and the second light-transmitting conductive layer is the same layer as that of the overlapping electrode.
8. The detection device according to claim 1, wherein a width in the first direction of the first metal layer is equal to a width in the first direction of the opening.
9. The detection device according to claim 4, further comprising signal line coupling wiring that is electrically coupled to the output signal lines, and extends in a second direction intersecting the first direction, wherein the first metal layer is provided in the same layer as that of the signal line coupling wiring, is coupled to the signal line coupling wiring, and is electrically coupled to the second metal layer through a first contact portion provided through the first interlayer insulating film.
10. The detection device according to claim 1, further comprising: a drive circuit that is provided in the peripheral region; and a plurality of second terminals that are provided adjacent in the first direction to the first terminals, and are electrically coupled to the drive circuit.
11. The detection device according to claim 10, further comprising a second interlayer insulating film that is stacked between the first interlayer insulating film and the insulating film in a region between the adjacent second terminals.
12. The detection device according to claim 10, further comprising outer edge wiring that is electrically coupled to each of at least more than one of the second terminals among the first terminals and the second terminals, and extends in a second direction intersecting the first direction between the second terminals and an outer periphery of the substrate.
13. The detection device according to claim 12, wherein the first interlayer insulating film extends in the second direction so as to cover the outer edge wiring, and a plurality of convex portions in each of which the outer edge wiring and the first interlayer insulating film are stacked are arranged in the first direction.
14. The detection device according to claim 13, wherein the outer edge wiring is not coupled to the first terminals, the detection device comprises: a plurality of first convex portions that are provided in a region between the first terminals and the outer periphery of the substrate, and are formed of the first interlayer insulating film; and a plurality of second convex portions that are provided in a

region between the second terminals and the outer periphery of the substrate, and in each of which the outer edge wiring and the first interlayer insulating film covering the outer edge wiring are stacked, and the first convex portions and the second convex portions are arranged in the first direction.